

Rafik El Khoury

AI Engineer for the Built Environment

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Summary

Civil engineer, urban planner, and AI researcher exploring the role of AI in the Architecture, Engineering, and Construction industry. Currently at IAAC building tools for the built environment — climate-aware festival shade placement, automated compliance checking, and project-manager communication pipelines. Owned the code in every university team project; **3rd place, Infrared.city Buildathon 2026** (solo); currently designing internal software for Rafik El-Khoury & Partners. Looking for a team where I can take ideas from concept to product — open to internships and collaborations. Authorized to work in the **UK** (Graduate Visa), **EU** (Greek citizenship), and **UAE** (Golden Visa).

Projects

FestCOOL / U-Shade — Climate-Aware Festival Shade Placement 2026

Decision-support system that places shade on the cells of a festival site that are simultaneously the hottest and the busiest — a real UTCI heat grid (infrared.city) crossed with a JuPedSim pedestrian-dynamics crowd field, solved as a Maximal Covering Location Problem, and clad in the lowest-transport-carbon reclaimed material the surrounding city already contains. Demoed live on **Primavera Sound 2026** at Parc del Fòrum, Barcelona.

- Demand field = crowd density $\times$ thermal excess above 26 °C UTCI on a 6 m lattice; crowd migrates on the real festival schedule with Spotify-grounded stage popularity. Placement solved twice — deterministic greedy as a permanent lower-bound witness, plus an exact integer program emitting a verifiable **optimality certificate**.
- Honesty as an engineered property — every number carries a provenance tier (grounded / derived / assumed / declared); the cooling-acceptance gate is hash-locked and fails loudly if thresholds are edited; the LLM only parses the brief (schema-constrained structured output, retry loop terminated on stop_reason) — it never designs.
- Stack — Python/FastAPI, infrared.city UTCI, JuPedSim, MCLP/ILP, Mapbox GL, React + TanStack Start, Anthropic Claude structured output. Team of 4 at IAAC MaAI01.

GigAI — Material Change Coordinator for Construction PMs 2026

Event-driven workflow automation that captures construction material changes from meetings and PM tools, then generates coordinated, PM-approvable actions with confidence scoring.

- Pipeline — Fireflies transcripts $\rightarrow$ event normalization (Claude Haiku) $\rightarrow$ context enrichment (pgvector RAG over floor plans, suppliers, history) $\rightarrow$ proposal generation (Claude Sonnet) $\rightarrow$ ranked actions for PM approval.
- Integrated Autodesk Construction Cloud, Google Calendar, Gmail; React+Vite dashboard with SSE for real-time decisions. 4-factor confidence scoring lands demo proposals at $\sim 85\%$.
- Stack — Python/FastAPI, React+Vite, PostgreSQL+pgvector, Docker, GCP Pub/Sub, Anthropic Claude (Haiku + Sonnet), Voyage AI embeddings.

ARCHAI — Automated Compliance Checking for Urban Regulations 2025 – 2026

Rhino-integrated tool that checks building geometry against urban planning regulations using vector-store retrieval and LLM-generated explanations.

- Pipeline — Rhino building geometry $\rightarrow$ Chroma vector-store rule retrieval (zone, metric, operator, threshold, source) $\rightarrow$ per-rule pass/fail table with article-cited LLM explanations.
- Tested against city regulation corpora (Mumbai, Barcelona) with a consistent rule-schema so the engine is regulation-agnostic.
- Stack — Python, ChromaDB (RAG), OpenAI GPT-4o-mini, Pandas, n8n; Rhino interop via CSV manifests.

CoolSpend — Street-Tree Cooling Optimiser (3rd Place, Infrared.city Buildathon 2026)

2026

Solo build. Tells Barcelona where to plant street trees, measures the cooling on live UTCI, and spends a fixed budget where heat hurts most.

- Four real datasets (satellite heat + imperviousness, OSM geometry, tree inventory, population register) fused into a 494-cell vulnerability grid; budgeted weighted max-coverage over a 4 m validated-slot lattice, cost-benefit greedy with the classical $(1 - 1/e)$ guarantee, ecology gate with zero invasive species.
- Every cooling figure is the measured ΔUTCI between two live Infrared.city runs (baseline vs planted) — Plaça dels Àngels 28 trees $\rightarrow$ 4,268 m² at €30/m²; citywide €1M $\rightarrow$ 90 trees, 6 sites, 20,609 m² cooled, 26,745 residents served.
- Stack — Python/FastAPI, numpy, shapely, pyproj, React 18 + TypeScript, deck.gl, mapbox-gl.

Experience

Graduate Urban Planner, Rafik El-Khoury & Partners — Consulting Engineers – Abu Dhabi, UAE Jan 2025 – Sept 2025

- Evaluated international masterplan proposals for a 19 km² island development; built a scoring framework weighting design merit, consultant expertise, and long-term vision alignment.
- Contributed to SHA villa complex redesign in Al-Jurf — piling works and peer review, CAD refinement, and real-estate sales brochure design.
- Programmed three community parks in Madinat Al-Riyadh through site analysis and public space planning.
- Authored UNICEF-guided school rehabilitation proposals in southern Lebanon.
- Briefed the president of Aramoun municipality on the town's urban character, planning issues, and proposed interventions.

Site Engineer, IMKAN Properties – United Arab Emirates Jan 2023 – Mar 2023

- Conducted villa inspections across multiple construction stages and produced weekly progress reports for all villas.
- Worked across construction sequences and methods, including tiling techniques and waterproofing applications.

Engineering Intern, Rafik El-Khoury & Partners — Consulting Engineers – Beirut, Lebanon Aug 2020 – Sept 2020

- Reviewed the Solid Waste Master Plan for Greater Beirut — assessed waste generation, environmental impact, and recycling strategies for sustainability and regulatory compliance.
- Supervised on-site activities at the Burj Hammoud Landfill — leachate collection oversight, waste-load recording, and worker safety procedure enforcement.

Education

Institute for Advanced Architecture of Catalonia (IAAC), MS in AI in Architecture and the Built Environment – Barcelona, Spain Oct 2025 – present

- Expected graduation July 2026.

University of Dundee, MSc in Spatial Planning with Sustainable Urban Design – Dundee, Scotland Sept 2023 – Sept 2024

- Team-based research-by-design masterplan for a new residential complex in Dundee's urban-rural fringe.

Quinta do Vale, Cert in Permaculture Design – Portugal July 2022 – Aug 2022

Loughborough University, BSc in Civil Engineering – Loughborough, UK Oct 2018 – June 2022

- Dissertation on Natural Flood Management — biodiversity and flood resilience in England.

Skills

AI/ML & Programming: Python, JavaScript/TypeScript, SQL; LLM APIs (OpenAI, Anthropic, Gemini), RAG (ChromaDB, pgvector), Voyage AI embeddings, Neo4j + OWL/Turtle knowledge graphs, NSGA-II multi-objective optimization (pymoo)

Agentic Development: Claude Code orchestration — 200+ agents and 150+ skills used strategically; deliberate, architecture-first AI-paired coding with full control of design decisions; schema-constrained structured output; led the codebase in every university team project

Backend / Frontend: FastAPI, Flask, React+Vite, PostgreSQL, Docker, Google Cloud Pub/Sub, n8n, Server-Sent Events, Git

Design & Engineering: Rhino, Grasshopper, AutoCAD, BIM, Ladybug environmental analysis, CFD integration (Infrared.city), IFC parsing

Planning & GIS: GIS, spatial analysis, Open Data Barcelona (CKAN), EPSG coordinate systems, statutory planning, report writing

Languages: English (fluent), French (fluent), Arabic (native), Spanish (conversational)